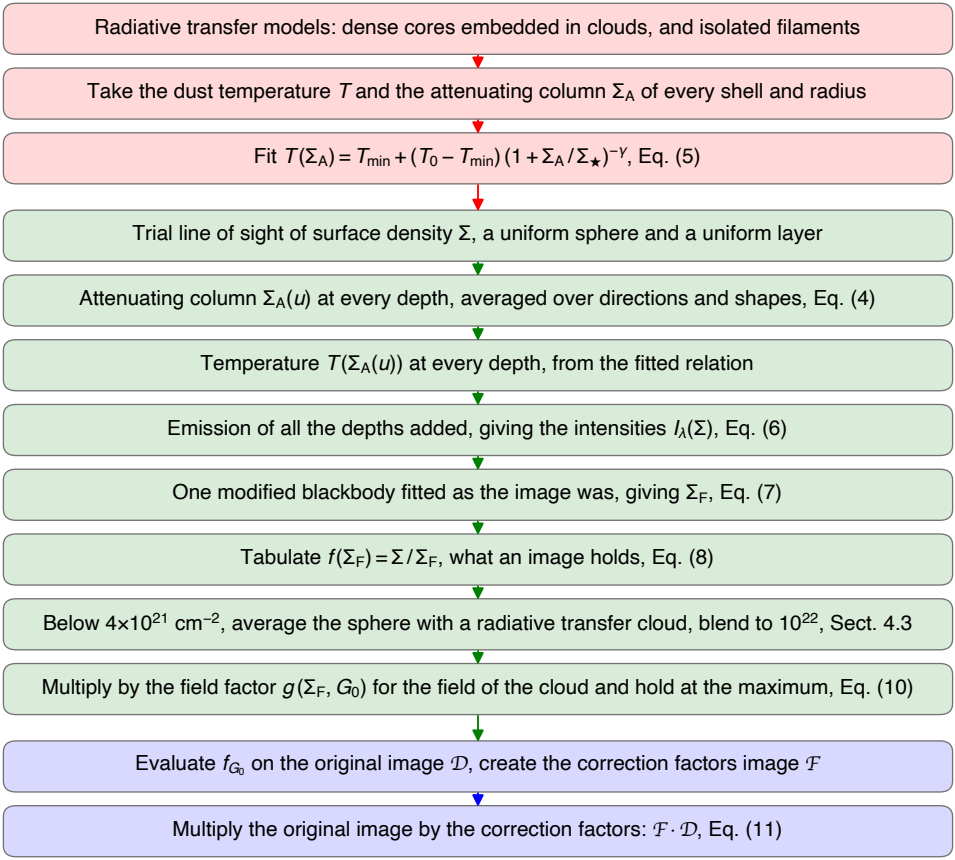
Radiative transfer models: dense cores embedded in clouds, and isolated filaments
Take the dust temperature T and the attenuating column Σ_A of every shell and radius
Fit $T(\Sigma_A) = T_{min} + (T_0 - T_{min})\,(1 + \Sigma_A / \Sigma_\star)^{-\gamma}$, Eq. (5)
Trial line of sight of surface density Σ, a uniform sphere and a uniform layer
Attenuating column $\Sigma_A(u)$ at every depth, averaged over directions and shapes, Eq. (4)
Temperature $T(\Sigma_A(u))$ at every depth, from the fitted relation
Emission of all the depths added, giving the intensities $I_\lambda(\Sigma)$, Eq. (6)
One modified blackbody fitted as the image was, giving Σ_F, Eq. (7)
Tabulate $f(\Sigma_F) = \Sigma / \Sigma_F$, what an image holds, Eq. (8)
Below $4{\times}10^{21}$ cm^{-2}, average the sphere with a radiative transfer cloud, blend to 10^{22}, Sect. 4.3
Multiply by the field factor $g(\Sigma_F, G_0)$ for the field of the cloud and hold at the maximum, Eq. (10)
Evaluate f_{G_0} on the original image $\mathcal{D}$, create the correction factors image $\mathcal{F}$
Multiply the original image by the correction factors: $\mathcal{F} \cdot \mathcal{D}$, Eq. (11)